

ML PROJECT IMPORTANT ATTRIBUTE

Based on your expanded problem statement, your system has two clear objectives: **predicting traffic congestion levels** and **estimating road clearance times**.

To handle this multi-objective scope (incorporating public events and historical patterns alongside your previous schema), you need a third table to bridge everything together.

Here is how you should structure your database and target variables to perfectly match your problem statement:

Table 1: Incidents (The Dynamic Event Log)

*This table tracks short-term road disruptions. It provides the data required to train the **Road Clearance Time** model.*

- `incident_id` (Primary Key)
- `incident_type` (Accident, Construction, Hazard, Stall)
- `severity_level` (Low, Medium, High)
- `vehicles_involved` (Integer)
- `lane_blockage` (None, Partial, Full)
- `timestamp_reported` (DateTime)
- `response_time_mins` (Integer)
- `weather_condition` (Text — e.g., Rainy, Clear, Snowy)
- `location_id` (Foreign Key — links to Table 2)
- `clearance_time_mins` (**Target Variable 1** — The label your clearance model will predict)

Table 2: Locations (The Static Roadway Catalog)

This table stores the fixed spatial assets of your road network.

- `location_id` (Primary Key)
- `road_name` (Text)
- `road_type` (Highway, Urban, Residential)
- `latitude` (Float)
- `longitude` (Float)

Table 3: Traffic History & Events (The Core Congestion Log)

*This new table is critical for your first objective: **Predicting future traffic congestion levels**. It records continuous hourly snapshots of the road network and cross-references them with public events.*

- `traffic_log_id` (Primary Key)
- `location_id` (Foreign Key — links to Table 2)
- `timestamp` (DateTime — logs historical patterns by hour/day)
- `public_event_type` (Text — e.g., Stadium Sports, Concert, Protest, None)
- `expected_event_attendance` (Integer — e.g., 50,000 people vs. 0)

- **active_incident_flag** (Binary — 1 if an active row exists in Table 1 for this hour, 0 if no)
- **congestion_level** (**Target Variable 2** — Scaled value like 0–5 or categorical like *Low, Medium, Heavy* that your congestion model will predict)